roamie

THE PROBLEM

Planning a trip requires using multiple apps while managing everything yourself.
and the Painpoints?

Too many apps, zero coordination
Flights, hotels, maps, all open, none connected.

Options everywhere, decisions nowhere
No app just tells you what to book.

One delay, everything breaks
A cancelled flight ruins your entire plan instantly.

OUR SOLUTION – WHAT WE'RE PROGRESSING TOWARD

🧠Deep Personalization Engine

- Capturing user preferences, budget, and behavior patterns
- Continuously adapting itineraries based on user choices and feedback
- Delivering context-aware travel plans, not one-time static outputs

🎯 Intelligent & EfficientSystem Design

- Building a unified multi-agent workflow where each component handles a specific task (planning, optimization, monitoring)
- Ensuring faster, more accurate outputs by reducing overload on a single model
- Creating a structured, decision-driven system instead of generic responses

⚡Offline-First AI Capability

- Integrating Ollama for on-device processing
- Enabling core features to function without internet access
- Providing privacy, low latency, and reliability during travel

BUT WHY ROAMIE ?

Context-Aware : Understandsand remembers theentire trip, not just one-time queries

Action-Oriented: Plans, optimizes, and updates instead of just suggesting

Real-World Ready : Works with budget, preferences, and live changes for practical travel plans

PROBLEM IMPACT

Why this matters — and why it's a massive market

300M+
Passengers disrupted every single year

2–4 hrs
Lostper incident rebooking manually

$800B
Global travel tech market

5 apps
Average used per trip none connected

$35B
Corporate travel management segment

72%
Travellers say disruptions ruin their experience

How the solution solves and to what extent

PS Requirement	Roamie Feature	Status
Detect travel disruptions	DisruptionCoordinator pipeline	√ Built
Find alternative flights	SearchAgent — AI-scored options	√ Built
Update itinerary automatically	ClaudeItineraryService rebuild	√ Built
Notify the traveller	ClawbotAgent via Claude Sonnet	√ Built
Proactive nearby suggestions	LiveSuggestionAgent	√ Built
Location + time awareness	ContextAnalyzerAgent	√ Built
Free time gap detection	Itinerary gap detector	√ Built
Trip purpose personalisation	Business / leisure preference layer	√ Built

FUTURE SCOPE & SCALING

Fourphasesfrom hackathon demo tomarketleader

PHASE 1 — 0–3 months

Real API Integration

- Amadeus API for live flight inventory
- Flightradar24 for real disruption detection
- Google Places for live nearby suggestions
- Stripe integration for booking flow

PHASE 2 — 3–6 months

B2B Enterprise Launch

- White-label API for travel management companies
- Corporate HR platform integrations
- SLA-backed disruption resolution guarantee
- Admin dashboard for travel managers

PHASE 3 — 6–12 months

Data & Personalisation Moat

- Disruption pattern learning from resolved events
- Preference fingerprinting across trips
- Predictive itinerary adjustments before disruption
- Expand to 20+ cities with local curation

PHASE 4 — 12–18 months

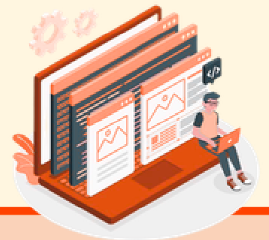
Ambient Travel Intelligence

- Proactive pre-trip risk alerting
- Voice-driven disruption resolution
- Airline SDK for in-app embedding
- Series A: $6–8M at $20–25M valuation

TECH STACK

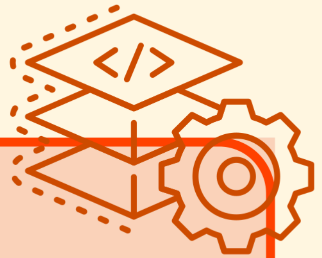
FRONTEND

- React 18 + TypeScript
- OpenStreetMap integration
- Multi-agent UI wiring



BACKEND

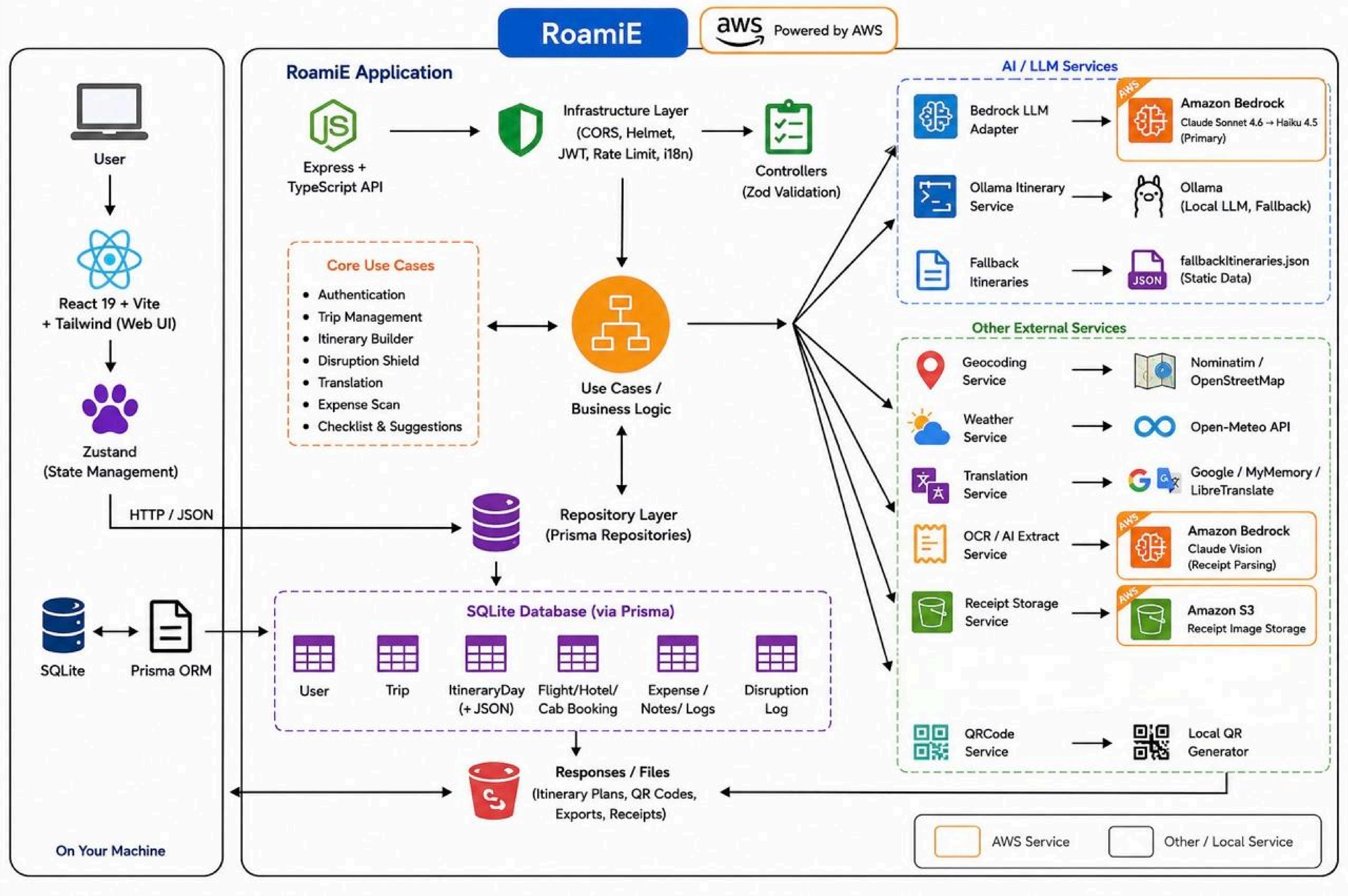
- Node.js / FastAPI (Python)
- Flask API



AI LAYER

- AWS Bedrock
- AWS S3
- AWS Strands
- Ollama for offline LLM
- PostgresAI

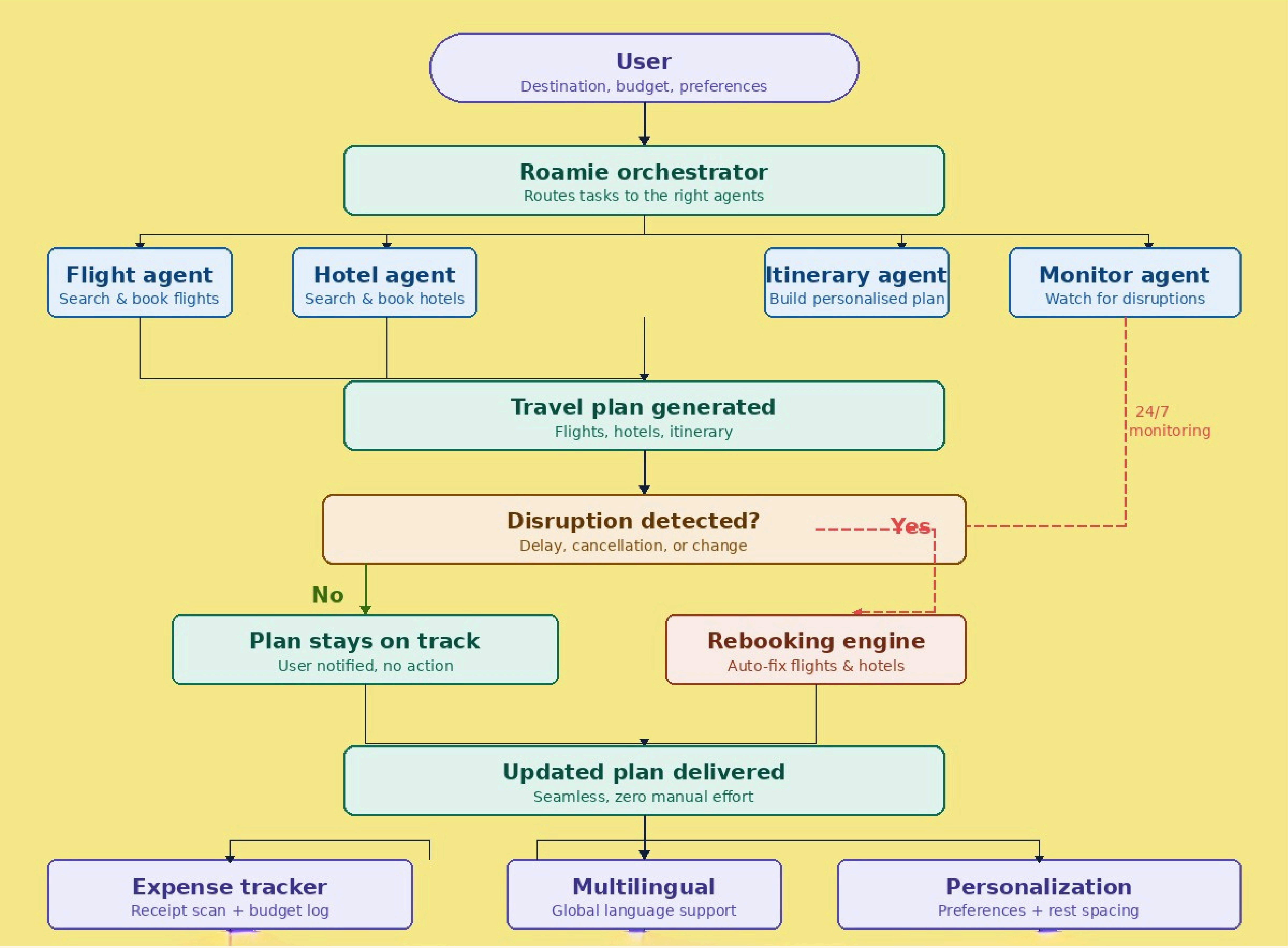
Built using Kiro and AIDLC


RoamiE
aws Powered by AWS
RoamiE Application
User
React 19 + Vite + Tailwind (Web UI)
Zustand (State Management)
HTTP / JSON
SQLite
Prisma ORM
On Your Machine
Express + TypeScript API
Infrastructure Layer (CORS, Helmet, JWT, Rate Limit, i18n)
Controllers (Zod Validation)
Core Use Cases
Authentication
Trip Management
Itinerary Builder
Disruption Shield
Translation
Expense Scan
Checklist & Suggestions
Use Cases / Business Logic
Repository Layer (Prisma Repositories)
SQLite Database (via Prisma)
User
Trip
ItineraryDay (+ JSON)
Flight/Hotel/ Cab Booking
Expense / Notes/ Logs
Disruption Log
Responses / Files (Itinerary Plans, QR Codes, Exports, Receipts)
AI / LLM Services
Bedrock LLM Adapter
AWS
Amazon Bedrock
Claude Sonnet 4.6 → Haiku 4.5 (Primary)
Ollama Itinerary Service
Ollama (Local LLM, Fallback)
Fallback Itineraries
JSON
fallbackItineraries.json (Static Data)
Other External Services
Geocoding Service
Nominatim / OpenStreetMap
Weather Service
Open-Meteo API
Translation Service
Google / MyMemory / LibreTranslate
OCR / AI Extract Service
AWS
Amazon Bedrock Claude Vision (Receipt Parsing)
Receipt Storage Service
AWS
Amazon S3 Receipt Image Storage
QRCode Service
Local QR Generator
AWS Service
Other / Local Service

WORKFLOW

Why our workflow is better

- **One system replaces five apps — search, book, plan, monitor, all in one.**
- **It doesn't just alert you — it fixes the problem automatically.**
- **24/7 monitoring means your plan adapts before you even notice the disruption.**



PROGRESS: CHECKPOINT 1 → FINAL SOLUTION

CHECKPOINT 1 — DELIVERED

- Complete UI/UX prototype with full navigation
- Conversational trip planning interface
- Budget-aware itinerary generation (multi-currency)
- Preference-based personalisation (adventure, food, culture)
- Interactive map visualisation for routes
- Dynamic day-wise itineraries with cost breakdown
- Responsive component-based architecture
- Expense AI scanner — receipt paste & categorise

→

FINAL SOLUTION — BUILT

- Multi-agent DisruptionCoordinator pipeline
- SearchAgent + BookingAgent + ClawbotAgent
- Claude Sonnet + Ollama dual AI adapter
- Live weather (Open-Meteo) + geocoding APIs
- 5-language i18n with real-time switching
- Smart packing list + cultural nudges
- Full Clean Architecture — swappable AI layer
- SSE streaming progress events to frontend

PROJECT DEPLOYMENT

Frontend → Amazon S3 + CloudFront

↓

Amazon API Gateway

↓

AWS Lambda

↓

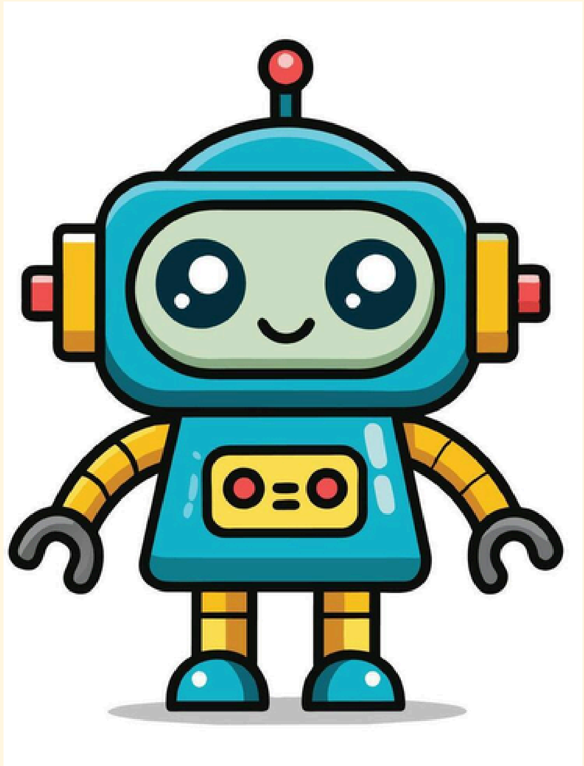
AWS Bedrock

↓

DynamoDB / RDS

1. Who is the customer?

- International leisure/ business travelers
- Book their own trips
- Comfortable using apps
- Faces flight disruptions, language barriers, unfamiliar local rules



2. Prevailing problem?

Trip planning is scattered across many apps, and breaks down the moment something changes — a delayed flight, an unfamiliar rule, a language barrier — leaving the traveler to manually fix it mid-trip.

3. Solution & key benefit?

One app: AI builds the itinerary, then watches and auto-fixes it when things go wrong

4. How do we describe it to customers?

AI travel companion that handles disruptions for you

5. How do we test & measure success?

Test with real travelers on trips with layovers/connections (higher disruption risk)

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We panic so you don't have to.

Thank you